09. Adnexal Masses: Initial Evaluation and Malignancy Risk

Companion reading handout for simultaneous listening.
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An adnexal mass is not a diagnosis.

It is a location.

That is the first discipline in this topic. The mass may be ovarian, tubal, paratubal, uterine, gastrointestinal, urinary, retroperitoneal, benign, malignant, infectious, pregnancy-related, or torsion. If the resident names it too quickly, the evaluation narrows too early.

The clinical challenge is to avoid two opposite errors.

One error is under-calling malignancy risk and allowing a patient who needs gynecologic oncology care to undergo the wrong operation in the wrong setting.

The other error is over-calling danger, ordering the wrong first tests, and sending a patient with benign disease into unnecessary surgery or unnecessary fear.

The dominant model is this: ultrasound morphology is the first risk language, interpreted through patient context.

The patient context tells you the pretest probability.

The ultrasound architecture tells you what kind of mass you are seeing.

The serum markers and risk tools refine the picture, but they do not replace the picture.

Start with the first triage question: is this an acute problem?

Acute unilateral pelvic pain in a reproductive-aged patient may be a hemorrhagic cyst. Intermittent unilateral pain that suddenly worsens may be torsion. Progressive pelvic pain with fever, chills, vomiting, and vaginal discharge suggests infection, especially tubo-ovarian abscess. A positive pregnancy test with pain and an adnexal finding raises ectopic pregnancy until proven otherwise.

Those are different pathways. The torsion patient does not wait for a leisurely malignancy workup. The pregnant patient needs the early-pregnancy algorithm. The patient with fever and discharge needs infection evaluation. The stable incidental mass needs careful risk stratification.

So the first resident reflex is: before asking, "Is it cancer?" ask, "Is she unstable, pregnant, infected, torsed, or safe for outpatient risk stratification?"

Patient context

Age is the most important independent ovarian cancer risk factor in the general population. Risk rises sharply after menopause. In the surveillance data cited by ACOG, the median age at ovarian cancer diagnosis was sixty three, and about sixty nine percent of patients were fifty five or older.

That does not mean every postmenopausal mass is cancer. Most adnexal masses in postmenopausal women are still benign neoplasms, such as cystadenomas. But the baseline risk is higher than in premenopausal women. That is why the same ultrasound finding or marker result can mean something different before and after menopause.

Family history is the other major context variable.

A strong family history of breast or ovarian cancer is not a small detail at the end of the history. It can change the entire risk frame.

ACOG gives a useful contrast. In the general population example, lifetime ovarian cancer risk is about one point six percent. For a thirty-five-year-old woman with one affected family member, the lifetime probability rises to about five percent. But for a woman with a BRCA one mutation, the lifetime risk of ovarian, fallopian tube, or peritoneal cancer is approximately forty one to forty six percent by age seventy. For BRCA two, it is ten to twenty seven percent. For Lynch syndrome, it is about five to ten percent through age seventy.

Those numbers are not trivia. They explain why family history belongs near the beginning of adnexal mass evaluation. Hereditary risk changes the meaning of an otherwise similar mass.

Symptoms

Symptoms do not diagnose the mass, but they route the differential.

Acute onset pain can be hemorrhagic cyst.

Unilateral intermittent pain that becomes acutely worse can be torsion.

Progressive pain with fever, chills, vomiting, and discharge can be tubo-ovarian abscess.

Dysmenorrhea and dyspareunia can suggest endometrioma.

Persistent bloating, generalized abdominal pain, and early satiety raise concern for malignancy.

Abnormal uterine bleeding or postmenopausal bleeding with a solid pelvic mass should make you think about estrogen-producing sex cord-stromal tumors, such as granulosa cell tumors.

That is the history pattern: route the differential without pretending history alone proves the diagnosis.

Physical examination has the same role. It contributes risk information, but it does not replace imaging.

Start with vital signs and general appearance. Examine lymph nodes, including cervical, supraclavicular, axillary, and groin nodes when malignancy is a concern. Examine the lungs, abdomen, and pelvis. Perform rectovaginal examination when indicated.

Concerning pelvic findings include a mass that is irregular, firm, fixed, nodular, bilateral, or associated with ascites.

But there is a trap here. Benign disease can mimic concerning findings. Endometriosis, chronic pelvic infection, hemorrhagic corpus luteum, tubo-ovarian abscess, and uterine leiomyomas can all produce confusing examination findings.

So the exam should raise or lower concern. It should not be treated as a final classifier.

Imaging

Transvaginal ultrasonography is the recommended initial imaging modality for a suspected or incidentally identified adnexal mass. ACOG says that no alternative imaging modality is superior enough to justify routine replacement of transvaginal ultrasound.

This is a major resident habit to install.

Do not jump to CT for initial characterization of an adnexal mass.

CT, MRI, and PET are not recommended for the initial evaluation. MRI can help when the origin of a pelvic mass is unclear, especially when the question is whether the mass is ovarian or a leiomyoma. CT is useful when cancer is suspected and the question becomes metastatic disease: ascites, omental metastases, peritoneal implants, pelvic or periaortic lymph node enlargement, hepatic metastases, obstructive uropathy, or another primary cancer such as pancreatic or colon cancer.

So the contrast is this.

Ultrasound characterizes the adnexal mass.

CT stages suspected cancer.

MRI solves selected origin or tissue-characterization questions.

PET is not the first step.

Abdominal ultrasound can be added when pelvic anatomy is distorted, when the mass extends beyond the pelvis, or when transvaginal ultrasound cannot be performed.

Now what should the ultrasound report actually describe?

It should describe size, cystic versus solid or mixed composition, laterality, septations, mural nodules, papillary excrescences, free fluid, and Doppler vascularity.

That list is the grammar of adnexal mass risk. If the report does not give you that grammar, it may not be enough to make a good decision.

The malignancy architecture

Ultrasound findings that raise concern include cyst size greater than ten centimeters, papillary or solid components, irregularity, ascites, and high color Doppler flow.

Notice the Doppler language. The concern is high color Doppler flow.

The exam trap is "high resistance." ACOG emphasizes high color Doppler flow as the concerning feature. Doppler indices can overlap between benign and malignant masses, so the resident should not overinterpret a single resistance index as if it were a cancer diagnosis.

The benign architecture

Benign features include a simple appearance, thin smooth walls, no solid components, no septations, and no internal blood flow on color Doppler.

This is the opposite mental image from malignancy architecture. A simple sonolucent cyst with smooth thin walls and no solid vascular tissue is a very different risk object from an irregular vascular mass with papillary projections and ascites.

Simple cysts are almost universally benign in any age group, with malignancy rates in most series of zero to one percent. Simple cysts up to ten centimeters may be safely monitored with repeat imaging when the transvaginal ultrasound was performed by experienced ultrasonographers, even in postmenopausal patients.

That statement is important because it prevents size from becoming the only thing the resident remembers.

Size matters, but architecture matters more.

A ten-centimeter irregular solid vascular mass with ascites is not the same as a well-characterized simple cyst. The number ten centimeters should increase attention, not erase morphology.

ACOG cites a large prospective study of two thousand seven hundred sixty three postmenopausal women with unilocular cysts no larger than ten centimeters followed with serial transvaginal ultrasound every six months. More than two thirds resolved spontaneously, and no cancers were detected after a mean follow-up of six point three years.

That does not mean every cyst can be ignored. It means morphology, expertise, and follow-up matter.

Some benign entities also have recognizable patterns.

An endometrioma often appears as a round homogeneous cyst with low-level internal echoes.

A mature teratoma may have echogenic and shadowing components.

A hydrosalpinx often appears as a tubular sonolucent cystic structure.

These patterns can help, but they still require correlation with symptoms, age, and persistence.

IOTA and formal ultrasound tools

The International Ovarian Tumor Analysis group developed structured tools, including Logistic Regression model two and Simple Rules, to estimate malignancy risk before surgery. A systematic review found IOTA Logistic Regression model two at a ten percent cutoff and IOTA Simple Rules among the best-performing models, with pooled sensitivities around zero point nine two to zero point nine three and specificities around zero point eight one to zero point eight three.

That performance is impressive.

But the resident should hear the caution clearly. These tools depend on ultrasound expertise, correct feature recognition, and validation context. Saying "IOTA" is not a substitute for understanding the morphology. The tool is only as good as the image interpretation placed into it.

So the reflex is: use structured tools when they fit the setting, but never let the name of the tool replace the architecture of the mass.

Serum markers

The most familiar marker is CA 125. It is useful, but it is also one of the easiest ways to mislead yourself.

CA 125 is most useful in postmenopausal women and for nonmucinous epithelial ovarian cancer. It is elevated in about eighty percent of epithelial ovarian cancers, but only about fifty percent of stage one epithelial ovarian cancers.

So a normal value does not rule out early cancer.

It is also not specific. CA 125 can rise with endometriosis, pregnancy, pelvic inflammatory disease, uterine leiomyomas, ascites of any cause, systemic lupus erythematosus, inflammatory bowel disease, and nongynecologic cancers.

So an elevated value does not automatically prove ovarian cancer.

This is why menopausal status changes the interpretation.

In a postmenopausal woman, the combination of an elevated CA 125 and a pelvic mass is highly suspicious for malignancy and should prompt referral to, or treatment in consultation with, a gynecologic oncologist.

In a premenopausal woman, CA 125 is less specific. Normal or mildly elevated values often occur with benign disease. A markedly elevated value raises concern, but even benign endometriomas can produce values of one thousand units per milliliter or greater.

Older ACOG guidance used a CA 125 threshold greater than two hundred units per milliliter for referral of a premenopausal woman with an adnexal mass. But ACOG notes that this was expert opinion and that no evidence-based threshold is currently available. The CA 125 value must be integrated with the ultrasound, exam, symptoms, age, and risk factors.

That is the key sentence.

CA 125 is a context amplifier, not a universal cancer detector.

Other serum markers are targeted to specific histologies.

If a germ cell tumor is suspected, beta hCG, alpha fetoprotein, and LDH may assist evaluation. In adolescents and young patients, this matters because germ cell tumors are relatively more important than epithelial ovarian cancer.

If a granulosa cell tumor is suspected, inhibin can help. The clinical pattern is a solid pelvic mass with irregular bleeding or postmenopausal bleeding, because granulosa cell tumors can produce estrogen.

Biomarker panels and multimodal tools

The multivariate index assay, the Risk of Ovarian Malignancy Algorithm, and the risk of malignancy index are designed to help decide who should be referred to or managed with a gynecologic oncologist, especially when a mass requires surgery.

But they are not recommended for use in the initial evaluation of an adnexal mass.

That distinction matters. Do not start with a panel because you do not yet know what the mass is. Start with context, exam, and ultrasound. Then use markers and formal tools when they answer the next question: does this patient need oncology involvement?

Referral

Consultation with or referral to a gynecologic oncologist is recommended when a postmenopausal patient has an adnexal mass plus elevated CA 125, ultrasound findings suggestive of malignancy, ascites, a nodular or fixed pelvic mass, or evidence of abdominal or distant metastasis.

For a premenopausal patient, referral or consultation is recommended with a very elevated CA 125, suspicious ultrasound findings, ascites, a nodular or fixed pelvic mass, or evidence of abdominal or distant metastasis.

Referral is also recommended when a formal risk assessment test is elevated, such as the multivariate index assay, the risk of malignancy index, the Risk of Ovarian Malignancy Algorithm, or an IOTA-based score.

Why does referral timing matter?

Because ovarian cancer outcomes are better when care is managed by physicians trained in ovarian cancer staging and debulking. Proper staging identifies occult metastasis, and aggressive debulking matters in advanced disease. ACOG notes that seventy five to eighty percent of women with ovarian cancer have advanced disease.

So the resident reflex is: refer before the wrong operation, not after it.

If a suspicious or persistent complex mass requires surgery, the operation should be performed by someone trained to stage and debulk ovarian cancer, in a facility with the support services needed, including pathology support such as frozen section when appropriate.

Now one more "do not do this first" topic: aspiration.

Aspiration of an adnexal mass for diagnosis is generally contraindicated when cancer is possible.

There are two reasons.

First, cytology sensitivity is limited. ACOG cites mixed results, with sensitivity only about fifty to seventy four percent. A negative aspiration does not safely exclude malignancy.

Second, aspiration of a malignant mass can spill and seed malignant cells into the peritoneal cavity, advancing stage and potentially worsening prognosis.

Even for a benign simple cyst, aspiration is often not definitive therapy. Recurrence by six months has been reported around forty four percent in premenopausal women and twenty five percent in postmenopausal women in one series, and thirty nine percent in another.

There are exceptions. Aspiration or drainage may be appropriate for tubo-ovarian abscess in selected cases, although antibiotics are first-line. Aspiration may also establish diagnosis in suspected advanced ovarian cancer when the patient is medically unfit for surgery and neoadjuvant chemotherapy is being considered.

But the basic resident reflex remains: do not aspirate a mass when cancer is still possible.

Run the model through cases

Case one.

A healthy twenty-four-year-old has pelvic pain and an adnexal mass. Before CT, before tumor-marker panels, and before anyone says "ovarian cyst" with confidence, confirm pregnancy status when indicated and obtain transvaginal ultrasound if appropriate. If she is virginal, prepubertal, or cannot undergo transvaginal imaging, transabdominal ultrasound may be the better route.

The goal is to identify whether this is pregnancy-related, hemorrhagic, torsion-risk, infectious, likely benign, or concerning.

Case two.

A thirty-two-year-old has a known endometrioma pattern on ultrasound and a CA 125 of eight hundred.

That number is uncomfortable. It should not be ignored. But in a premenopausal patient, endometriosis can markedly elevate CA 125, even to one thousand units per milliliter or greater. The decision should integrate morphology, symptoms, exam, persistence, and risk factors. CA 125 alone should not create a cancer diagnosis.

Case three.

A sixty-eight-year-old has a pelvic mass and elevated CA 125.

That is a different risk world. In a postmenopausal woman, elevated CA 125 plus a pelvic mass is highly suspicious and should trigger gynecologic oncology referral or consultation.

Case four.

The ultrasound shows a simple unilocular cyst, thin smooth wall, no solid component, no septations, no internal flow, and size below ten centimeters. It was performed by an experienced ultrasonographer.

That architecture is reassuring. Even in a postmenopausal patient, this kind of simple cyst may be monitored with repeat imaging rather than immediate surgery, if the whole clinical picture fits.

Case five.

The ultrasound shows papillary projections, solid vascular areas, irregularity, ascites, and high color Doppler flow.

That architecture escalates. This is not a "watch it casually" mass. This is malignancy-risk architecture, and referral or consultation should occur before surgical planning goes down the wrong path.

The final synthesis

An adnexal mass is a location, not a diagnosis.

The initial evaluation sorts the patient into acute danger, pregnancy-related, infectious, likely benign, uncertain, or malignancy-risk pathways.

Transvaginal ultrasound is the first-line characterization test. Describe the architecture: size, cystic or solid composition, septations, nodules, papillary excrescences, free fluid, and vascularity.

Simple, thin-walled, avascular cysts reassure. Papillary, solid, irregular, ascitic, high-flow masses escalate.

CA 125 is strongest after menopause and weakest when used as a stand-alone cancer detector before menopause. Biomarker panels and formal scores can refine referral decisions when surgery is being considered, but they are not the first evaluation.

And the final resident reflex is this.

Describe architecture before management. Interpret markers through menopause and morphology. Refer before the wrong operation, not after it.